MATH 3032 — Mathematical Statistics

Chapter 3: Point Estimation

Complete Exam Preparation Guide

All 8 Exam Questions · Worked Examples · Practice Problems · Mock Exam

Topics covered

Q1	Likelihood function $L(\theta)$
Q2	MLE $\hat{\theta}$ + second derivative test
Q3	Bias via $\mathbb{E}[X_1]$ and u-substitution
Q4	Mean Squared Error
Q5	Consistency (LLN and Chebyshev)
Q6	Fisher information $I(\theta)$
Q7	Cramér-Rao lower bound
Q8	Asymptotic normality

Matthew Setiadi

Accelerated B.S./M.S. Computational Data Science

Temple University

March 20, 2026

Textbooks:
Panaretos, *Statistics for Mathematicians* (Ch. 3)
Hogg, Tanis & Zimmerman, *Probability and Statistical Inference*, 9th ed. (§6.4, §6.6)

MATH 3032 — Mathematical Statistics

Chapter 3: Point Estimation — Complete Exam Prep

All 8 Exam Questions · Worked Examples · Practice Problems · Mock Exam

Textbooks: Panaretos, *Statistics for Mathematicians* (Ch. 3) & Hogg–Tanis–Zimmerman, *Probability and Statistical Inference*, 9th ed. (§6.4, §6.6)

Exam structure: 8 questions on a single continuous pdf, 30 points total. Questions 1–2 are $\approx \frac{1}{3}$ of points; Questions 3–5 are $\approx \frac{1}{3}$; Questions 6–8 are $\approx \frac{1}{3}$.

Contents

The Statistical Framework

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} F_\theta$ with pdf $f(x;\theta)$, $\theta \in \Theta \subset \mathbb{R}$ unknown. The goal of Chapter 3 is to:

1. Construct an estimator $\hat{\theta}$ — a function of the sample that approximates θ.
2. Analyse its quality: bias, mean squared error, consistency.
3. Assess optimality: does $\hat{\theta}$ achieve the Cramér-Rao lower bound?
4. Describe its large-sample distribution.

$$\int_0^\infty e^{-u}\,du = 1, \qquad \int_0^\infty u\,e^{-u}\,du = 1, \qquad \int_0^\infty u^2 e^{-u}\,du = 2$$
$$\int_0^\infty u^3 e^{-u}\,du = 6, \qquad \int_0^\infty u^4 e^{-u}\,du = 24, \qquad \int_0^\infty u^n e^{-u}\,du = n!$$

Also: $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$; $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$; $\mathrm{Var}(\sum X_i) = n\,\mathrm{Var}(X_1)$ if i.i.d.

The Eight Exam Questions — Theory

Q2.1. Likelihood Function $L(\theta)$

Given a pdf $f(x;\theta)$, write the likelihood function $L(\theta)$ for an i.i.d. sample $X_1, \dots, X_n$ and simplify.

Definition. The likelihood function is the joint pdf of the sample, viewed as a function of θ:

$$L(\theta) = \prod_{i=1}^n f(X_i;\theta).$$

Since the samples are i.i.d., the joint pdf factors into a product of marginals. The key simplification steps are:

1. Write out all n factors explicitly.
2. Group **constants** (terms involving only θ): $\prod c(\theta) = c(\theta)^n$.
3. Group **data terms** (products of functions of X_i): $\prod g(X_i)$.
4. **Collapse exponentials**: $\prod e^{h(X_i,\theta)} = \exp\big(\sum h(X_i,\theta)\big)$.

Always identify the *exponent's structure* in $f(x;\theta)$ first. If the exponent is $-x/\theta$, then L will contain $e^{-\sum X_i/\theta}$. If the exponent is $-x^2/\theta^2$, then L will contain $e^{-\sum X_i^2/\theta^2}$. This exponent determines which sufficient statistic appears in $\hat{\theta}$.

Review-session worked example. Let $f(x;\theta) = \frac{x^2 e^{-x/\theta}}{2\theta^3}$, $x > 0$. Then:

$$L(\theta) = \prod_{i=1}^{n} \frac{X_i^2\, e^{-X_i/\theta}}{2\theta^3} = \underbrace{\frac{1}{(2\theta^3)^n}}_{\text{constant}^n} \cdot \underbrace{\prod_{i=1}^{n} X_i^2}_{\text{data}} \cdot \underbrace{\exp\Big(-\frac{\sum X_i}{\theta}\Big)}_{\text{collapsed exp.}}$$

Q2.2. Maximum Likelihood Estimator $\hat{\theta}$

Find $\hat{\theta}_{\text{MLE}} = \arg\max_\theta L(\theta)$. **Justify that it is a maximum** using the second derivative test.

Method.

1. Take the **log-likelihood** $\ell(\theta) = \log L(\theta)$. Since log is strictly increasing, $\arg\max L = \arg\max \ell$.
2. **Differentiate** $\ell(\theta)$ w.r.t. θ and set $\ell'(\theta) = 0$ (score equation).
3. **Solve** for $\hat{\theta}$.
4. **Second derivative test**: evaluate $\ell''(\hat{\theta})$. If $\ell''(\hat{\theta}) < 0$, then $\hat{\theta}$ is a local maximum.

After finding $\hat{\theta}$, the score equation $\ell'(\hat{\theta}) = 0$ gives a relation of the form $\sum(\text{data}) = n \cdot g(\hat{\theta})$ for some function g. Substitute this into $\ell''(\hat{\theta})$ to simplify the expression and determine its sign without knowing the data values explicitly.

Review-session worked example. Continuing from Q1:

$$\ell(\theta) = -n\log(2\theta^3) + 2\sum \log X_i - \frac{\sum X_i}{\theta}$$

$$\ell'(\theta) = -\frac{3n}{\theta} + \frac{\sum X_i}{\theta^2} = 0 \implies \hat{\theta} = \frac{\sum X_i}{3n} = \frac{\bar{X}}{3}.$$

Second derivative test:

$$\ell''(\theta) = \frac{3n}{\theta^2} - \frac{2\sum X_i}{\theta^3}.$$

From the score equation: $\sum X_i = 3n\hat{\theta}$. Substituting:

$$\ell''(\hat{\theta}) = \frac{3n}{\hat{\theta}^2} - \frac{2 \cdot 3n\hat{\theta}}{\hat{\theta}^3} = \frac{3n}{\hat{\theta}^2} - \frac{6n}{\hat{\theta}^2} = -\frac{3n}{\hat{\theta}^2} < 0. \quad \checkmark$$

Q2.3. Bias of $\hat{\theta}$

Compute $\text{Bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta}] - \theta$. This requires evaluating $\mathbb{E}[X_1]$ by integration.

Method.

1. Use **linearity** to reduce $\mathbb{E}[\hat{\theta}]$ to $\mathbb{E}[X_1]$: e.g. if $\hat{\theta} = \frac{1}{cn}\sum X_i$, then $\mathbb{E}[\hat{\theta}] = \frac{\mathbb{E}[X_1]}{c}$.
2. Compute $\mathbb{E}[X_1] = \int_0^\infty x\, f(x;\theta)\, dx$ via **u-substitution** $u = x/\theta$ (or $u = x^2/\theta^2$ for Weibull-type pdfs).

3. Apply the gamma integral formula from the formula sheet.
4. Compute Bias $= \mathbb{E}[\hat{\theta}] - \theta$.

Review-session worked example. With $f(x;\theta) = \frac{x^2 e^{-x/\theta}}{2\theta^3}$ and $\hat{\theta} = \bar{X}/3$:

$$\mathbb{E}[X_1] = \int_0^\infty x \cdot \frac{x^2 e^{-x/\theta}}{2\theta^3}\,dx = \frac{1}{2\theta^3}\int_0^\infty x^3 e^{-x/\theta}\,dx.$$

Substitute $u = x/\theta$, so $x = \theta u$, $dx = \theta\,du$:

$$= \frac{1}{2\theta^3} \cdot \theta^4 \int_0^\infty u^3 e^{-u}\,du = \frac{\theta}{2} \cdot 3! = \frac{\theta}{2} \cdot 6 = 3\theta.$$

Therefore $\mathbb{E}[\hat{\theta}] = \frac{3\theta}{3} = \theta$, so $\text{Bias}(\hat{\theta},\theta) = 0$. $\hat{\theta}$ is **unbiased**.

For Gamma-type pdfs with pdf $\propto x^{\alpha-1}e^{-x/\theta}$: the integral $\mathbb{E}[X_1]$ uses the $(n = \alpha)$ row of the table, and $\mathbb{E}[X_1^2]$ uses the $(n = \alpha+1)$ row. In general, $\mathbb{E}[X_1^k]$ uses row $n = \alpha + k - 1$, giving value $(\alpha + k - 1)!$. For Weibull-type pdfs with e^{-x^2/θ^2} in the exponent, use $u = x^2/\theta^2$ instead.

Q2.4. Mean Squared Error

Compute $\text{MSE}(\hat{\theta},\theta) = \mathbb{E}[(\hat{\theta}-\theta)^2]$.

Decomposition (Panaretos Remark 3.5):

$$\text{MSE}(\hat{\theta},\theta) = \text{Bias}(\hat{\theta},\theta)^2 + \text{Var}(\hat{\theta}).$$

If Bias $= 0$, then MSE $= \text{Var}(\hat{\theta})$.

Method for computing $\text{Var}(\hat{\theta})$:

1. Pull the constant out squared: $\text{Var}\big(\frac{1}{cn}\sum X_i\big) = \frac{1}{c^2n^2} \cdot n\,\text{Var}(X_1) = \frac{\text{Var}(X_1)}{c^2 n}$.
2. Compute $\text{Var}(X_1) = \mathbb{E}[X_1^2] - (\mathbb{E}[X_1])^2$.
3. $\mathbb{E}[X_1^2]$ requires the same u-substitution, using the next row of the gamma table.

The professor's explicit rule: your MSE *must* have n in the denominator. If it doesn't, you dropped the n somewhere in the variance-of-sum step: $\text{Var}(\sum X_i) = n\,\text{Var}(X_1)$, so $\text{Var}(\hat{\theta}) = \text{Var}(X_1)/(c^2 n)$, not $\text{Var}(X_1)/c^2$.

Review-session worked example. With $\mathbb{E}[X_1] = 3\theta$:

$$\begin{aligned}
\mathbb{E}[X_1^2] &= \frac{1}{2\theta^3}\int_0^\infty x^4 e^{-x/\theta}\,dx = \frac{\theta^2}{2}\int_0^\infty u^4 e^{-u}\,du = \frac{\theta^2}{2} \cdot 4! = \frac{\theta^2}{2} \cdot 24 = 12\theta^2.\\
\text{Var}(X_1) &= 12\theta^2 - (3\theta)^2 = 12\theta^2 - 9\theta^2 = 3\theta^2.\\
\text{Var}(\hat{\theta}) &= \frac{1}{(3n)^2} \cdot n \cdot 3\theta^2 = \frac{3n\theta^2}{9n^2} = \frac{\theta^2}{3n}.\\
\text{MSE}(\hat{\theta},\theta) &= 0 + \frac{\theta^2}{3n} = \boxed{\frac{\theta^2}{3n}}.
\end{aligned}$$

Q2.5. Consistency

Show that $\hat{\theta} \xrightarrow{p} \theta$ as $n \to \infty$.

There are two approaches; choose based on $\hat{\theta}$'s structure.

Approach 1 — Law of Large Numbers. If $\hat{\theta} = \frac{1}{n}\sum g(X_i)$:

$$\text{By LLN:} \quad \frac{1}{n}\sum g(X_i) \xrightarrow{p} \mathbb{E}[g(X_1)] = \theta. \quad \checkmark$$

Requires: $\mathbb{E}[|g(X_1)|] < \infty$ (verified since we computed $\mathbb{E}[X_1] < \infty$).

Approach 2 — Chebyshev inequality. Always valid when MSE is computed:

$$P(|\hat{\theta} - \theta| \geq \varepsilon) \leq \frac{\mathrm{MSE}(\hat{\theta}, \theta)}{\varepsilon^2} = \frac{\theta^2}{3n\varepsilon^2} \to 0 \text{ as } n \to \infty.$$

By the squeeze theorem, $\hat{\theta} \xrightarrow{p} \theta$. ✓

LLN: use when $\hat{\theta}$ is visibly $(1/n)\sum g(X_i)$. Cleaner write-up.
Chebyshev: universal fallback — use whenever MSE is already computed (i.e. Q4 is done). Also handles cases where $\hat{\theta}$ is not a simple average (e.g. $\hat{\theta} = \sqrt{\sum X_i^2/n}$).
Key insight (Panaretos Remark 3.8): $\mathrm{MSE}(\hat{\theta}, \theta) \to 0 \Rightarrow \hat{\theta} \xrightarrow{p} \theta$.

Q2.6. Fisher Information $I(\theta)$

Compute $I(\theta) = \mathbb{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(X_1;\theta)\right)^2\right]$.

Method.

1. Compute $\log f(x;\theta)$ — this is the $n = 1$ case of the log-likelihood.
2. Differentiate in θ to get the **score** $s(\theta) = \frac{\partial}{\partial\theta}\log f(X_1;\theta)$.
3. Compute $I(\theta) = \mathbb{E}[s(\theta)^2]$.

Under regularity conditions, $\mathbb{E}[s(\theta)] = 0$, so $I(\theta) = \mathbb{E}[s^2] = \mathrm{Var}(s)$. If the score takes the form

$$s(\theta) = \frac{X_1 - \mathbb{E}[X_1]}{c(\theta)},$$

then immediately $I(\theta) = \mathrm{Var}(X_1)/c(\theta)^2$ — no new integrals needed. This pattern appears whenever the log-likelihood is linear in X_1.

Alternative formula (negative second derivative):

$$I(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log f(X_1;\theta)\right].$$

Review-session worked example.

$$\log f = -\log(2\theta^3) + 2\log X_1 - \frac{X_1}{\theta}, \qquad s(\theta) = \frac{X_1}{\theta^2} - \frac{3}{\theta} = \frac{X_1 - 3\theta}{\theta^2}.$$

Since $\mathbb{E}[X_1] = 3\theta$, the score is $s(\theta) = (X_1 - \mathbb{E}[X_1])/\theta^2$. Therefore:

$$I(\theta) = \frac{\mathrm{Var}(X_1)}{\theta^4} = \frac{3\theta^2}{\theta^4} = \frac{3}{\theta^2}.$$

Q2.7. Cramér-Rao Lower Bound

Does $\hat{\theta}_{\mathrm{MLE}}$ attain the Cramér-Rao lower bound?

Theorem 3.9 (Cramér-Rao, Panaretos). Let T be any estimator with bias $\beta(\theta) = \mathbb{E}[T] - \theta$. Under regularity conditions:

$$\mathrm{Var}(T) \geq \frac{(1 + \beta'(\theta))^2}{n \cdot I(\theta)}.$$

For **unbiased** estimators ($\beta = 0$, $\beta' = 0$):

$$\mathrm{Var}(T) \geq \frac{1}{n \cdot I(\theta)} =: \mathrm{CRLB}.$$

For estimating a function $g(\theta)$ (e.g. $g(\theta) = \theta^2$):

$$\mathrm{Var}(T) \geq \frac{[g'(\theta)]^2}{n \cdot I(\theta)}.$$

Review-session worked example.

$$\mathrm{CRLB} = \frac{1}{n \cdot I(\theta)} = \frac{1}{n \cdot 3/\theta^2} = \frac{\theta^2}{3n}.$$

Since $\mathrm{Var}(\hat{\theta}) = \frac{\theta^2}{3n} = \mathrm{CRLB}$, the MLE **attains the Cramér-Rao bound**. Therefore $\hat{\theta}$ is the UMVUE — no unbiased estimator has smaller variance.

For $f(x;\theta) = 1/\theta$, $0 < x < \theta$: the support depends on θ, violating a regularity condition. The score $s(\theta) = -1/\theta$ contains no X_1, so $\mathbb{E}[s] \neq 0$. The MLE is $\hat{\theta} = X_{(n)} = \max(X_1, \ldots, X_n)$, which is biased: $\mathbb{E}[X_{(n)}] = n\theta/(n+1)$. The bias-corrected estimator $T^* = \frac{n+1}{n} X_{(n)}$ has $\mathrm{Var}(T^*) = \theta^2/(n(n+2))$, which is *strictly less than* the formal CRLB θ^2/n — demonstrating that the bound is invalid for this model.

Q2.8. Asymptotic Normality

Find σ such that $\dfrac{\sqrt{n}}{\sigma}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1)$.

Approach 1 — Classical CLT. If $\hat{\theta} = \frac{1}{n}\sum g(X_i)$:

$$\frac{\sqrt{n}\,(\hat{\theta} - \theta)}{\sqrt{\mathrm{Var}(g(X_1))}} \xrightarrow{d} N(0,1), \qquad \sigma = \sqrt{\mathrm{Var}(g(X_1))}.$$

Approach 2 — MLE CLT (Panaretos §3.3.3). For any regular MLE:

$$\sqrt{n \cdot I(\theta)}\,(\hat{\theta} - \theta) \xrightarrow{d} N(0,1), \qquad \sigma = \frac{1}{\sqrt{I(\theta)}}.$$

Both give $\sigma^2 = 1/I(\theta)$, and both equal $\mathrm{Var}(g(X_1))$, only when:

$$\mathrm{Var}(g(X_1)) = \frac{1}{I(\theta)} \quad\Longleftrightarrow\quad \mathrm{Var}(\hat{\theta}) = \frac{1}{n \cdot I(\theta)} = \text{CRLB}.$$

This three-way identity $\mathrm{Var}(g(X_1)) = 1/I(\theta) = n \cdot \mathrm{Var}(\hat{\theta})$ holds *if and only if* the MLE attains the Cramér-Rao bound. When the bound is not attained, the two approaches give different values of σ, and the MLE CLT (Approach 2) may not apply in its standard form.

Review-session worked example.

$$\text{Approach 1:}\quad \hat{\theta} = \tfrac{1}{n}\textstyle\sum(X_i/3), \quad \sigma = \sqrt{\mathrm{Var}(X_1/3)} = \sqrt{\tfrac{3\theta^2}{9}} = \tfrac{\theta}{\sqrt{3}}.$$

$$\text{Approach 2:}\quad \sigma = \frac{1}{\sqrt{I(\theta)}} = \frac{1}{\sqrt{3/\theta^2}} = \frac{\theta}{\sqrt{3}}.$$

$\boxed{\sigma = \theta/\sqrt{3}}$. Both agree, confirming the CR bound is attained.

Worked Practice Problems

Practice 1: Weibull$(2, \theta)$ — Likelihood

Distribution. $X_1, \dots, X_n \overset{\text{iid}}{\sim} f(x;\theta) = \frac{2x}{\theta^2} e^{-x^2/\theta^2}$, $x > 0$, $\theta > 0$.

Key structural observation. The exponent is $-x^2/\theta^2$ (not $-x/\theta$). This means: (1) the collapsed exponential in L will be $e^{-\sum X_i^2/\theta^2}$, and (2) the correct u-substitution is $u = x^2/\theta^2$.

Likelihood:

$$L(\theta) = \prod_{i=1}^{n} \frac{2X_i}{\theta^2} e^{-X_i^2/\theta^2} = \frac{2^n}{\theta^{2n}} \cdot \prod_{i=1}^{n} X_i \cdot \exp\!\Big(-\frac{\sum X_i^2}{\theta^2}\Big).$$

Practice 2: Rayleigh(θ) — MLE and Second Derivative

Distribution. $f(x;\theta) = \frac{2x}{\theta} e^{-x^2/\theta}$, $x > 0$. (Note: parameter θ appears as θ, not θ^2, in the exponent — watch the powers carefully.)

Log-likelihood:

$$\ell(\theta) = n \log 2 - n \log \theta + \sum \log X_i - \frac{\sum X_i^2}{\theta}.$$

Score equation $\ell'(\theta) = 0$:

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{\sum X_i^2}{\theta^2} = 0 \quad \Longrightarrow \quad \hat{\theta} = \frac{\sum X_i^2}{n}.$$

Second derivative test:

$$\ell''(\theta) = \frac{n}{\theta^2} - \frac{2\sum X_i^2}{\theta^3}.$$

From the score equation: $\sum X_i^2 = n\hat{\theta}$. Therefore:

$$\ell''(\hat{\theta}) = \frac{n}{\hat{\theta}^2} - \frac{2n\hat{\theta}}{\hat{\theta}^3} = \frac{n}{\hat{\theta}^2} - \frac{2n}{\hat{\theta}^2} = -\frac{n}{\hat{\theta}^2} < 0. \quad \checkmark$$

Practice 3: Gamma$(3, \theta)$ — Full Q1–Q8 Chain

Distribution. $f(x;\theta) = \frac{x^2 e^{-x/\theta}}{2\theta^3}$, $x > 0$. (Gamma with shape $\alpha = 3$, scale θ.)

Quantity	**Computation**	**Result**
$L(\theta)$	$\prod f(X_i;\theta)$	$(2\theta^3)^{-n}\prod X_i^2\cdot e^{-\sum X_i/\theta}$
$\hat\theta$	$\ell'(\theta)=0$	$\bar X/3=\sum X_i/(3n)$
$\ell''(\hat\theta)$	sub $\sum X_i=3n\hat\theta$	$-3n/\hat\theta^2<0$ ✓
$\mathbb{E}[X_1]$	$u=x/\theta$, row $n=3$	3θ
$\mathrm{Bias}(\hat\theta,\theta)$	$\mathbb{E}[\hat\theta]-\theta=\theta-\theta$	0 (unbiased)
$\mathbb{E}[X_1^2]$	$u=x/\theta$, row $n=4$	$12\theta^2$
$\mathrm{Var}(X_1)$	$12\theta^2-(3\theta)^2$	$3\theta^2$
$\mathrm{MSE}(\hat\theta,\theta)$	$\mathrm{Var}(X_1)/(9n)$	$\theta^2/(3n)$
Consistency	Chebyshev: $\theta^2/(3n\varepsilon^2)\to 0$	$\hat\theta\xrightarrow{p}\theta$
$I(\theta)$	$\mathrm{Var}(X_1)/\theta^4$	$3/\theta^2$
CRLB	$1/(nI(\theta))$	$\theta^2/(3n)$
Bound attained?	$\mathrm{Var}(\hat\theta)=\mathrm{CRLB}$	Yes — UMVUE
σ (Q8)	$1/\sqrt{I(\theta)}$	$\theta/\sqrt{3}$

Practice 4: Exponential(λ) — Fisher Information

Distribution. $f(x;\lambda)=\lambda\,e^{-\lambda x}$, $x>0$.

Score function:

$$\log f=\log\lambda-\lambda X_1,\qquad s(\lambda)=\frac{1}{\lambda}-X_1=-(X_1-\underbrace{\mathbb{E}[X_1]}_{=1/\lambda}).$$

Fisher information (using the variance shortcut):

$$I(\lambda)=\mathrm{Var}(X_1)=\frac{1}{\lambda^2}.$$

CRLB and σ:

$$\mathrm{CRLB}=\frac{1}{n\cdot 1/\lambda^2}=\frac{\lambda^2}{n},\qquad \sigma=\frac{1}{\sqrt{I(\lambda)}}=\lambda.$$

Practice 5: Uniform$(0,\theta)$ — Bound Not Attained

Distribution. $f(x;\theta)=1/\theta$, $0<x<\theta$. **Support depends on** θ — regularity conditions fail.

MLE. $L(\theta)=\theta^{-n}\mathbf{1}(X_{(n)}\le\theta)$ is decreasing in θ for $\theta\ge X_{(n)}$, so $\hat\theta=X_{(n)}$.

Bias. The pdf of $X_{(n)}$ is $f_{(n)}(x)=nx^{n-1}/\theta^n$. Then:

$$\mathbb{E}[X_{(n)}]=\frac{n}{\theta^n}\int_0^\theta x^n\,dx=\frac{n\theta}{n+1},\qquad \mathrm{Bias}=-\frac{\theta}{n+1}.$$

Bias-corrected estimator. $T^*=\frac{n+1}{n}X_{(n)}$ is unbiased. Its variance:

$$\mathrm{Var}(T^*)=\frac{\theta^2}{n(n+2)}.$$

Formal CRLB. With $s(\theta)=-1/\theta$ (constant in X_1!) and $I(\theta)=1/\theta^2$:

$$\mathrm{CRLB}=\frac{\theta^2}{n}.$$

Comparison.

$$\mathrm{Var}(T^*) = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{n} = \mathrm{CRLB}.$$

T^* **beats the formal CRLB** — possible only because the regularity conditions fail. The bound is not a valid lower bound for this model.

Consistency: Two Approaches Compared

Both approaches prove $\hat{\theta} \xrightarrow{p} \theta$ but differ in what they require.

	Law of Large Numbers	**Chebyshev / MSE**
Best when	$\hat{\theta} = (1/n)\sum g(X_i)$	MSE already computed (Q4 done)
Requires	$\mathbb{E}[g(X_1)] = \theta$; $\mathbb{E}[\lvert g(X_1)\rvert] < \infty$	$\text{MSE}(\hat{\theta}, \theta) \to 0$
Key step	LLN: $(1/n)\sum g(X_i) \xrightarrow{p} \mathbb{E}[g(X_1)] = \theta$	$P(\lvert\hat{\theta} - \theta\rvert \geq \varepsilon) \leq \text{MSE}/\varepsilon^2 \to 0$
Fails when	$\hat{\theta}$ not an average (e.g. $X_{(n)}$, ratio)	MSE doesn't vanish
Example	$\hat{\theta} = \bar{X}/3 = (1/n)\sum(X_i/3)$	$\hat{\theta} = \sqrt{\sum X_i^2/n}$

Panaretos Remark 3.8: $\text{MSE}(\hat{\theta}, \theta) \to 0$ implies $\hat{\theta} \xrightarrow{p} \theta$. Therefore if you've computed MSE in Q4 and it goes to zero, consistency (Q5) is essentially already proven.

The Three-Way Identity

When the CR bound is attained, the following quantities are all equal:

$$\boxed{\text{Var}(g(X_1)) \;=\; \frac{1}{I(\theta)} \;=\; n \cdot \text{Var}(\hat{\theta}) \;=\; n \cdot \text{CRLB} \;=\; \sigma^2}$$

This identity simultaneously explains why:

1. The classical CLT and the MLE CLT give the *same* σ for Q8.
2. The UMVUE achieves the exact lower bound.
3. $\hat{\theta}$ is an *efficient* estimator.

When the bound is *not* attained (e.g. Uniform$(0, \theta)$), the identity breaks, the two CLT routes diverge, and the standard MLE CLT no longer applies.

Mock Exam: Weibull$(2, \theta)$ — Full Q1–Q8

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x;\theta) = \frac{2x}{\theta^2} e^{-x^2/\theta^2}$, $x > 0$, $\theta > 0$. Work all eight questions. The formula sheet in Section 1 is available.

Q6.1. Likelihood $L(\theta)$

Write and simplify $L(\theta) = \prod_{i=1}^{n} f(X_i;\theta)$.

Solution.

$$L(\theta) = \frac{2^n}{\theta^{2n}} \cdot \prod_{i=1}^{n} X_i \cdot \exp\Bigl(-\frac{\sum X_i^2}{\theta^2}\Bigr).$$

Group: $(2/\theta^2)^n$ from the constant; $\prod X_i$ from the data; $\exp(-\sum X_i^2/\theta^2)$ from collapsing the exponentials.

Q6.2. MLE $\hat{\theta}$ and second derivative test

Solution.

$$\ell(\theta) = n\log 2 - 2n\log\theta + \sum \log X_i - \frac{\sum X_i^2}{\theta^2}.$$
$$\ell'(\theta) = -\frac{2n}{\theta} + \frac{2\sum X_i^2}{\theta^3} = 0 \quad\Longrightarrow\quad \hat{\theta} = \sqrt{\frac{\sum X_i^2}{n}}.$$
$$\ell''(\theta) = \frac{2n}{\theta^2} - \frac{6\sum X_i^2}{\theta^4}.$$

From score equation: $\sum X_i^2 = n\hat{\theta}^2$. Then: $\ell''(\hat{\theta}) = 2n/\hat{\theta}^2 - 6n/\hat{\theta}^2 = -4n/\hat{\theta}^2 < 0$. ✓

Q6.3. Bias via $\mathbb{E}[X_1^2]$

Solution. The natural unbiased estimator is $\tilde{\theta}^2 = \sum X_i^2/n$ for the target θ^2. Compute:

$$\mathbb{E}[X_1^2] = \frac{2}{\theta^2}\int_0^\infty x^3 e^{-x^2/\theta^2}\,dx.$$

Substitute $u = x^2/\theta^2$, so $x^2 = \theta^2 u$ and $2x\,dx = \theta^2\,du$, i.e. $x^3\,dx = \theta^2 u \cdot \frac{\theta^2}{2}\,du = \frac{\theta^4}{2}u\,du$:

$$\mathbb{E}[X_1^2] = \frac{2}{\theta^2}\cdot\frac{\theta^4}{2}\int_0^\infty u\,e^{-u}\,du = \theta^2 \cdot 1 = \theta^2.$$

Therefore $\mathbb{E}[\tilde{\theta}^2] = \theta^2$, so $\mathrm{Bias}(\tilde{\theta}^2, \theta^2) = 0$.

Q6.4. MSE of $\tilde{\theta}^2$

Solution.

$$\mathbb{E}[X_1^4] = \frac{2}{\theta^2}\int_0^\infty x^5 e^{-x^2/\theta^2}\,dx.$$

Same substitution: $x^5\,dx = (x^2)^2 x\,dx = \theta^4 u^2 \cdot \frac{\theta^2}{2}\,du = \frac{\theta^6}{2}u^2\,du$.

$$\mathbb{E}[X_1^4] = \frac{2}{\theta^2}\cdot\frac{\theta^6}{2}\cdot 2! = 2\theta^4.$$

$$\mathrm{Var}(X_1^2) = 2\theta^4 - (\theta^2)^2 = \theta^4, \qquad \mathrm{Var}(\tilde{\theta}^2) = \frac{\theta^4}{n}, \qquad \mathrm{MSE}(\tilde{\theta}^2, \theta^2) = \boxed{\frac{\theta^4}{n}}.$$

Q6.5. Consistency of $\tilde{\theta}^2$

Solution.

LLN: $\tilde{\theta}^2 = (1/n)\sum X_i^2 \xrightarrow{p} \mathbb{E}[X_1^2] = \theta^2$. ✓

Chebyshev: $P(|\tilde{\theta}^2 - \theta^2| \geq \varepsilon) \leq \theta^4/(n\varepsilon^2) \to 0$. ✓

Q6.6. Fisher Information $I(\theta)$

Solution.

$$\log f = \log 2 + \log x - 2\log\theta - \frac{x^2}{\theta^2}, \qquad s(\theta) = -\frac{2}{\theta} + \frac{2X_1^2}{\theta^3} = \frac{2}{\theta^3}(X_1^2 - \theta^2).$$

Since $\mathbb{E}[X_1^2] = \theta^2$, the score is $(2/\theta^3)(X_1^2 - \mathbb{E}[X_1^2])$:

$$I(\theta) = \frac{4}{\theta^6}\operatorname{Var}(X_1^2) = \frac{4}{\theta^6}\cdot\theta^4 = \frac{4}{\theta^2}.$$

Q6.7. Does $\tilde{\theta}^2$ attain the CR bound?

Solution. Estimating $g(\theta) = \theta^2$, so $g'(\theta) = 2\theta$:

$$\text{CRLB} = \frac{[g'(\theta)]^2}{n\cdot I(\theta)} = \frac{4\theta^2}{n\cdot 4/\theta^2} = \frac{\theta^4}{n}.$$

Since $\operatorname{Var}(\tilde{\theta}^2) = \theta^4/n = \text{CRLB}$, the bound **is attained**. $\tilde{\theta}^2$ is the UMVUE for θ^2.

Q6.8. Asymptotic Normality — find σ

Solution.
Approach 1 (Classical CLT): $\tilde{\theta}^2 = (1/n)\sum X_i^2$, so $Y_i = X_i^2$ are i.i.d. with $\operatorname{Var}(Y_1) = \theta^4$.

$$\sigma = \sqrt{\operatorname{Var}(X_1^2)} = \theta^2.$$

Approach 2 (MLE CLT): $\sigma = g'(\theta)/\sqrt{I(\theta)} = 2\theta/\sqrt{4/\theta^2} = 2\theta\cdot\theta/2 = \theta^2$.

Both agree: $\boxed{\sigma = \theta^2}$, consistent with the CR bound being attained.

Quick Reference Summary

Q1 — Likelihood: $L(\theta) = c(\theta)^n \cdot \prod g(X_i) \cdot \exp\left(-h(\sum X_i^k)/\theta^m\right)$

Q2 — MLE: $\ell'(\hat{\theta}) = 0$; substitute $\sum(\text{data}) = n \cdot g(\hat{\theta})$ into $\ell''(\hat{\theta})$ to show < 0.

Q3 — Bias: $\text{Bias} = \mathbb{E}[\hat{\theta}] - \theta$; reduce to $\mathbb{E}[X_1]$ via linearity; u-sub $u = x/\theta$ or $u = x^2/\theta^2$; use $\int_0^\infty u^n e^{-u}\, du = n!$

Q4 — MSE: $\text{MSE} = \text{Bias}^2 + \text{Var}(\hat{\theta})$; $\text{Var}(\hat{\theta}) = \text{Var}(X_1)/(c^2 n)$; need $\mathbb{E}[X_1^2]$ (one row higher in table).

Q5 — Consistency: LLN if $\hat{\theta} = (1/n)\sum g(X_i)$ and $\mathbb{E}[g(X_1)] = \theta$; or Chebyshev if MSE$\to 0$.

Q6 — Fisher info: $I(\theta) = \text{Var}(s(\theta))$ where $s = \frac{\partial}{\partial\theta}\log f(X_1;\theta)$; shortcut: if $s = (X_1 - \mathbb{E}[X_1])/c(\theta)$ then $I(\theta) = \text{Var}(X_1)/c(\theta)^2$.

Q7 — CR bound: $\text{CRLB} = 1/(nI(\theta))$ for unbiased estimators; $[g'(\theta)]^2/(nI(\theta))$ for estimating $g(\theta)$. Attained iff $\text{Var}(\hat{\theta}) =$CRLB. If support depends on θ: regularity fails, bound may be invalid.

Q8 — Asymptotic normality: Classical CLT: $\sigma = \sqrt{\text{Var}(g(X_1))}$. MLE CLT: $\sigma = 1/\sqrt{I(\theta)}$. These agree iff bound attained.

- ☐ $L(\theta)$: grouped correctly? Exponent collapsed to $\exp(-\sum X_i^k/\theta^m)$?
- ☐ MLE: second derivative test done and $\ell''(\hat{\theta}) < 0$ shown explicitly?
- ☐ Bias: u-substitution tracks all four pieces (substitution, x^k, $e^{-(\cdot)}$, dx)?
- ☐ MSE: n appears in the denominator?
- ☐ Consistency: stated the conclusion "$\hat{\theta} \xrightarrow{p} \theta$" explicitly?
- ☐ Fisher info: used $\mathbb{E}[s(\theta)] = 0$ so $I(\theta) = \text{Var}(s)$? No new integrals?
- ☐ CR bound: computed CRLB; stated whether attained; cited regularity if not?
- ☐ Q8: wrote both CLT approaches; confirmed they agree; stated σ clearly?